

Koketso Maseki

Data Analytics : SQL

Exercise 1

01. Selecting and Sorting

1. Retrieve all columns from the employee table

Expected columns: ID, first-name, last-name, department, salary, hire-date, city.

Syntax:

```
SELECT ID,  
       first-name,  
       last-name,  
       department,  
       salary,  
       hire-date,  
       city,  
FROM table - employee
```

ID	first-name	last-name	department	Salary	hire-date	City
1	Alice	Green	IT	70000	2020/01/10	Tobung
2	Brian	Lee	HR	45000	2019/03/22	Cape T
3	Cathy	Zulu	Finance	65000	2018/07/18	Durban
4	David	Mokoena	Marketing	50000	2021/11/05	Pretoria
5	Eva	Naidoo	IT	72000	2017/09/30	Tobung

2. Find all unique departments

Expected column: department

Syntax:

```
SELECT DISTINCT department,  
FROM employee-table;
```

Department
IT
HR
Finance
Marketing

3. Retrieve first and last name ordered by salary descending
Expected column: first-name, last-name

```
SELECT first-name,
       last-name,
FROM employee-table
ORDER BY salary DESC
```

first name	last name
Eva	Maidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. Retrieve the top 3 highest-paid employees.
Expected columns: ID, first-name, last-name, salary

```
SELECT id,
       first-name,
       last-name,
       salary,
FROM employee-table
```

ID	first name	last name	salary
5	Eva	Maidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

2. Filtering with WHERE

5. Find employees in the IT department

Expected columns: id, first_name, last_name, department

```
SELECT id,  
        last_name,  
        first_name,  
        department,  
FROM employee_table  
WHERE department IN ('IT')
```

ID	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. Find employees in Finance with salary > 60 000

Expected columns: ID, first_name, last_name, department

```
SELECT id,  
        first_name,  
        last_name,  
        department,  
FROM employees_table  
WHERE salary > 60 000 salary IN ('finance') > 60 000
```

ID	first_name	last_name	department	Salary
3	Cathy	Zulu	finance	65 000

7. Find employees in HR or Marketing

Expected columns: id, first_name, last_name, department

```
SELECT id,  
       first_name,  
       last_name,  
       department  
FROM employee_table  
WHERE department IN HR OR Marketing
```

ID	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. Find employees not in IT

Expected columns: id, first_name, last_name, department

```
SELECT id,  
       first_name,  
       last_name,  
       department  
FROM employee_table;  
WHERE department NOT IN IT
```

ID	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. Find employees in IT, HR or Finance using IN
 Expected columns: id, first_name, last_name, department

```
SELECT id,
       first_name,
       last_name,
       department,
FROM employees-table;
WHERE department IN ('IT', 'HR', 'Finance')
```

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Maidoo	IT

10. Find employees in IT with salary > 65 000 and city Johannesburg

Expected columns: id, first_name, last_name, department, salary, city

```
SELECT id,
       first_name,
       last_name,
       department,
```

```
SELECT *
FROM employees-table;
WHERE salary > 65 000
WHERE department IN ('IT')
      salary > 65 000 AND city =
      Johannesburg
```

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Maidoo	IT	72 000	Johannesburg